

Maram Sripathi

ENGINEERING

Java Programming

CO1-AT3 - CODING CHALLENGE

View Only

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QUESTIONS

Q1

Smart University Payroll Management System

25 marks

Q2

Smart Vehicle Insurance Claim System

25 marks

2/2 answered

Smart University Payroll Management System

25 marks

A university employs three categories of staff: Teaching Faculty, Technical Staff, and Visiting Faculty. The payroll system should calculate the monthly salary based on the employee type while maintaining a common employee hierarchy. Requirements Create a superclass Employee with the following private data members: employeeId, employeeName, department, basicSalary. Provide: Default constructor, Parameterized constructor, Getter and Setter methods, displayEmployee(), calculateSalary(). Create the following subclasses: TeachingFaculty, teachingHours, incentivePerHour, Salary: basicSalary + (teachingHours * incentivePerHour), TechnicalStaff, overtimeHours, overtimeRate, Salary: basicSalary + (overtimeHours * overtimeRate), VisitingFaculty, lecturesHandled, paymentPerLecture, Salary: lecturesHandled * paymentPerLecture. Override calculateSalary() in all subclasses. Create an array of Employee objects and demonstrate runtime polymorphism. Display: Employee details, Monthly salary, Highest-paid employee, Average salary, Search an employee using the employee ID.

Test Case 1 (Normal Case)

Input

3

TeachingFaculty

101

Dr. Arun

CSE

50000

20

500

TechnicalStaff

102

Mr. Ravi

ECE

35000

15

300

VisitingFaculty

103

Name: Mr. Ravi

Salary: 39500.0

Your Code

```
1 import java.util.*;
2
3
4 class Employee
5 {
6     private int employeeId;
7     private String employeeName;
8     private String department;
9     private double basicSalary;
10
11     public Employee()
12     {
13     }
14
15     public Employee(int employeeId, String employeeName, String department, double basicSalary)
16     {
17         this.employeeId = employeeId;
18         this.employeeName = employeeName;
19         this.department = department;
20         this.basicSalary = basicSalary;
21     }
22
23     public int getEmployeeId()
24     {
25         return employeeId;
26     }
27
28     public String getEmployeeName()
29     {
30         return employeeName;
31     }
32 }
```

Input

3

TeachingFaculty

101

Output

Employee Details
Employee ID : 101
Name : Dr. Arun
Department : CSE
Monthly Salary : 40000.0

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

0.00 / 25 marks

Q2 Smart Vehicle Insurance Claim System

25 marks

Problem Statement An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type. **Requirements** Superclass: Vehicle Private members: vehicleNumber ownerName insuredAmount Subclasses: Car Bike Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles Store them using an array of superclass references Calculate claim amounts using runtime polymorphism Search by vehicle number Display the vehicle with the highest claim amount **Question 3 (Very Hard): Online Shopping Order Processing System (50 Marks)** **Problem Statement** An e-commerce company sells products in three categories: Electronics Clothing Grocery The billing process varies for each category. **Requirements** Superclass: Product Private members: productId productName price quantity Subclasses: Electronics Clothing Grocery Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references Calculate the final bill using runtime polymorphism Search products by ID Display the most expensive order Compute total revenue

Claim Formula (Recommended)

Car

Claim = insuredAmount × 0.80

Bike

Claim = insuredAmount × 0.60

Truck

Claim = insuredAmount × 0.90

Test Case 1

Input

3

Car

TN10AB1234

Ramesh

800000

Your Code

```

1 import java.util.*;
2
3
4 abstract class Vehicle
5 {
6     private String vehicleNumber;
7     private String ownerName;
8     private double insuredAmount;
9
10    public Vehicle(String vehicleNumber, String ownerName, double insuredAmount)
11    {
12        this.vehicleNumber = vehicleNumber;
13        this.ownerName = ownerName;
14        this.insuredAmount = insuredAmount;
15    }
16
17    public String getVehicleNumber()
18    {
19        return vehicleNumber;
20    }
21    public String getOwnerName()
22    {
23        return ownerName;
24    }
25
26    public double getInsuredAmount()
27    {
28        return insuredAmount;
29    }
30
31    public abstract double calculateClaimAmount();
32
33    public static void main(String[] args)
34    {
35        Scanner sc = new Scanner(System.in);

```

Input

```

3
Car
TN10AB1234
Ramesh

```

Output

```

Vehicle : TN10AB1234
Claim Amount : 640000.0
Vehicle : TN09XY5678
Claim Amount : 480000.0
Vehicle : TN11TR9999
Claim Amount : 720000.0

```

Visible Test Cases

Test Case 1 ✖

Test Case 2 ✖

Test Case 3 ✖

0.00 / 25 marks